

Abdullah Shamout

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EDUCATION

Istanbul Technical University

Bachelor of Computer engineering

October 2021 – July 2025

Istanbul, Turkey

Inzva X Google Developers ML Bootcamp

TensorFlow Developer Certificate

June 2023 – October 2023

Istanbul, Turkey

EXPERIENCE

Artificial Intelligence and Software Engineer

Cezeri Robotics / Baykar Technologies

October 2024 – Present

Istanbul, Turkey

- Founded an internal product initiative for 3D scenery generation, leading the project from concept to development
- Held technical meetings for requirements/feedback with the client during development
- Migrate legacy products to modern frameworks, establishing professional and scalable environment setups
- Containerize applications and build robust CI/CD pipelines to streamline development and production workflows
- Improved a utilized Structure from Motion (SfM) algorithm speed by 14% through targeted optimizations
- Worked on 3D Scene Reconstruction using Volumetric and Radiance Field methods
- Utilized SOTA technologies such as Gaussian Splatting and NeRF based techniques
- Worked on utilizing LLMs as agentic mediators between sensory controls, actuators, and user panel
- Integrated agentic LLM models within report generation pipelines for live UAV flight communication with base
- Utilized technologies like RAG and Text2SQL within the developed products
- Contribute to full-stack software development, including frontend, backend, networking, and low-level systems

Research and Development Intern

Arçelik/Beko Corporate

August 2024 – September 2024

Istanbul, Turkey

- Created a dataset that relates the designs of TVs to their EMI signals using Anechoic Chambers
- Worked on creating an application to create the dataset automatically into a compact labeled form using tkinter
- Developed ML and DL models using Sklearn and TensorFlow for the prediction of EMI signals
- Dealt with multivariate series forecasting, dataset optimization, and big data analysis
- Dealt with data visualization and preprocessing using pandas and seaborn

Undergraduate Student Researcher

Istanbul Technical University's SiMiT Lab

November 2023 – July 2024

Istanbul, Turkey

- Developed several insect classification models using TensorFlow and PyTorch
- Worked with multi-label classification models that work on a taxonomic level
- Employed probabilistic prediction thresholds and utilized transfer learning
- Extensive literature review of SOTA employed technologies in the fields of object detection and classification

Software Development Intern

RealSoft Advanced Applications

July 2023 – September 2023

Amman, Jordan

- Developed a product supplier-retailer application with the mobile development team using flutter
- Developed some APIs for the supplier-retailer application using the .NET framework
- Executed structural database migrations in MySQL to improve schema efficiency and ensure data integrity.

NOTABLE PROJECTS

🌿 NeRF vs Gaussian Splatting | Python, PyTorch, Blender

June 2025

- Wrote the entire NeRF paper from scratch using python and pytorch
- Collected real live data using a drone
- Trained my NeRF model and the Gaussian Splatting paper model on my data
- Authored a comparative study analyzing the performance and trade-offs of both methods

🌿 StyleGAN & Pixel2Style2Pixel Animations | StyleGAN3, Google Colab, OpenCV, Pixel2Style2Pixel

May 2025

- Generated high-resolution face images using pretrained StyleGAN3.

- Interpolated latent vectors to animate realistic face morphing between identities.
- Used pSp encoder to convert real celebrity photos into GAN latent vectors.
- Created image-to-image morph animations and visualized transitions in high quality.

🔗 **Osteoid Scoliosis Treatment** | *Jittor, PyTorch, Blender, Typescript, Python* October 2024 – April 2025

- Developed a human body Non-Watertight, Non-Manifold mesh segmentation model to extract the human torso
- Developed a scoliosis classification model that maps the torso to a suitable brace for treatment based on its degree
- Developed a web-based user interface that streamlines the entire process and visualizes the processed meshes

🔗 **Inferno** | *Gazebo, ROS, RViz, Python, OpenCV* December 2024

- Implemented A* pathfinding with B-spline smoothing for efficient robot navigation.
- Developed reactive obstacle avoidance with Pure Pursuit for smooth trajectory tracking.
- Designed frontier-based exploration and vantage point selection for full map coverage.
- Built HSV-based flag detection using OpenCV for object recognition and retrieval.
- Integrated all modules in a ROS/Gazebo simulation for end-to-end autonomous operation.

🔗 **Gofies** | *Express JS, Flask, Docker Compose, Nginx, Github Workflows, Prometheus, Grafana, Loki, Promtail, Google Cloud Platform(GCP), PyTorch, Hugging Face, Project Management, MongoDB* October 2024 – December 2024

- Managed a team that develops a system that manages multiple hospitals, clinics, doctors, lab staff, and patients
- Contributed to the development team in DevOps, Cloud, AI, and Microservice backend engineering roles
- Managed and distributed tasks for each subteam
- Developed a Medical Multi-Modal Expert System as a Microservice
- Setup environments for all modules of the system and containerized them
- Orchestrated all containers and integrated them with scripts and CI/CD pipelines
- Setup Cloud environment with all needed metrics/logs
- Deployed application with a load balancing webserver

For a complete list of all projects and coursework, please visit my portfolio: 🔗 Portfolio
Note some of the projects are not publicly displayed and can be shown in private sessions

ACTIVITIES

DIOPSIS ARISE challenge participant March 2024

- Joined the DIOPSIS ARISE challenge as a participant representing Istanbul Technical University
- Created Detection and Multilevel classification models for insects in the challenge

Introduction to Programming with C — ACM Club Workshop February 2024

- Organized and taught introductory programming sessions in C for students through the ACM Student Chapter

English and Mathematics Tutor for Underprivileged High School Students September 2022 – June 2023

- Volunteered as a teacher, providing English and Mathematics instruction to high school students from underprivileged backgrounds
- Designed lesson plans and offered mentorship to support academic growth and confidence

Türkiye Bursları Scholarship Winner August 2021

- Full scholarship to study at Istanbul Technical University in Computer Engineering

Dean's List of Honours

- Awarded honours and high honours on the Dean's list for high finishing GPAs for multiple semesters

CERTIFICATES

Google TensorFlow Developer Certificate

Deep Learning Specialization(5 courses)

Machine Learning Specialization(3 courses)

DeepLearning.AI TensorFlow Developer(4 courses)

TECHNICAL SKILLS

Languages: Python, C/C++, C#, SQL, JavaScript, Typescript, Dart, HTML, CSS, Verilog, LaTeX, ARM Cortex M0+

Frameworks: TensorFlow, PyTorch, Jittor, Hugging face, ROS2, React, Vite, Tailwind, Flask, Flutter, .NET

Developer Tools: Git, Docker, Bash, ZSH, Nginx, GCP VS Code, Visual Studio, PyCharm, Eclipse, Vivado

LANGUAGES

English : Native/bilingual

Arabic : Native/bilingual

Turkish : C2 level/Fluent